



Sir Syed CASE Institute of Technology, Islamabad

Spring 2026

Assignment 01
Artificial Intelligence – AI 2101

Announced Date: March 17, 2026

Due Date: March 29, 2026

Total Marks = 20

Instructor Name: Hamid Ashfaq

Marks Obtained =

[CLO2, C3]

Water Jug Problem:

The water Jug Problem is a problem where two jugs of water are given, say one is a 4-litre one, and the other one is a 3-litre one, but none of the measuring markers is mentioned on any of it. There is a pump available to fill the jugs with water. How can you exactly pour 2 liters of water into a 4-litre jug? Assuming that both the jugs are empty, the task is to find a solution to pour 2-litre water into a 4-litre jug.

1. Formulate the problem (Write its states, initial state, actions, transition model, goal test and path cost).
2. Write the algorithm to solve this problem.
3. This problem can be solved using Breadth First Search and Depth First Search. Dry run the both uninformed searching techniques to find the solution. (Dry run on the page and add the screenshots in the word file)
4. Write the python code for this problem.

Note: Submit a pdf document having question 1 to 3 and 4th question code with Output screenshot. Also, ipynb file for the question 4.